Bangladesh University of Engineering and Technology
Department of Computer Science and Engineering

Course: CSE 206
Digital Logic Design Sessional

Experiment No. 3
Truth tables and K-maps

1. Simplify the following functions into a minimal SOP form using K-maps. Implement them using 7400-series IC chips.

$$f(A, B, C, D, E) = \sum(0, 6, 9, 11, 18, 20) + d(2, 4, 7, 8, 15, 16, 22)$$

2. Simplify the following functions into a minimal POS form using K-maps. Implement them using 7400-series IC chips.

$$f(A, B, C, D) = \prod(0, 2, 6, 8, 9, 10).d(5, 12, 13, 14, 15)$$

Lab Sheet : For each of the problems/questions, your lab sheet should cover:

- Truth Table
- K-map
- Simplification steps
- Required IC chips
- Circuit diagram

Answer the following questions:

1. Design and implement a circuit using gates that takes two 2-bit numbers as input and generates their product as its output.
2. Design a circuit that takes a 5-bit BCD number (00–19) as input and produces its binary equivalent as output.